

Training Log Interpreter

This notebook parses and visualizes training logs from the Int2Int model training. It extracts model parameters, task information, and plots metrics over epochs.

Configuration

Specify the path to your training log file:

Parse Training Log

Parsed 103 parameters and 226 epoch logs

Report Model Configuration and Training Parameters

```
=====
MODEL CONFIGURATION AND TRAINING PARAMETERS
=====
```

– EXPERIMENT INFO

Experiment Name: mu
Experiment ID: 1
Task Type: MU
Operation: data

– DATA

Training Data: /mnt/c/Users/ziwen/clair/mobius_case_study/input/input_dir_interCRT100_natural/mu_interCRT100_natural.txt.train
Eval Data: /mnt/c/Users/ziwen/clair/mobius_case_study/input/input_dir_interCRT100_natural/mu_interCRT100_natural.txt.test
Data Types: int[200]:range(-1,2)
Base: 1000
Modulus: 67

– MODEL ARCHITECTURE

Architecture: encoder_decoder
Encoder Layers: 4
Decoder Layers: 4
Encoder Embedding Dim: 256
Decoder Embedding Dim: 256
Encoder Heads: 8
Decoder Heads: 8
Dropout: 0.1
Attention Dropout: 0.1

– TRAINING PARAMETERS

Optimizer: adam_inverse_sqrt,lr=0.00025
Batch Size: 128
Eval Batch Size: 128
Epoch Size: 50000
Max Epochs: 201
Eval Size: 10000
Gradient Clipping: 5
Max Length: 512
Max Output Length: 512

– OTHER SETTINGS

FP16: False
CPU Mode: False
Multi-GPU: False
Num Workers: 0

```
=====
```

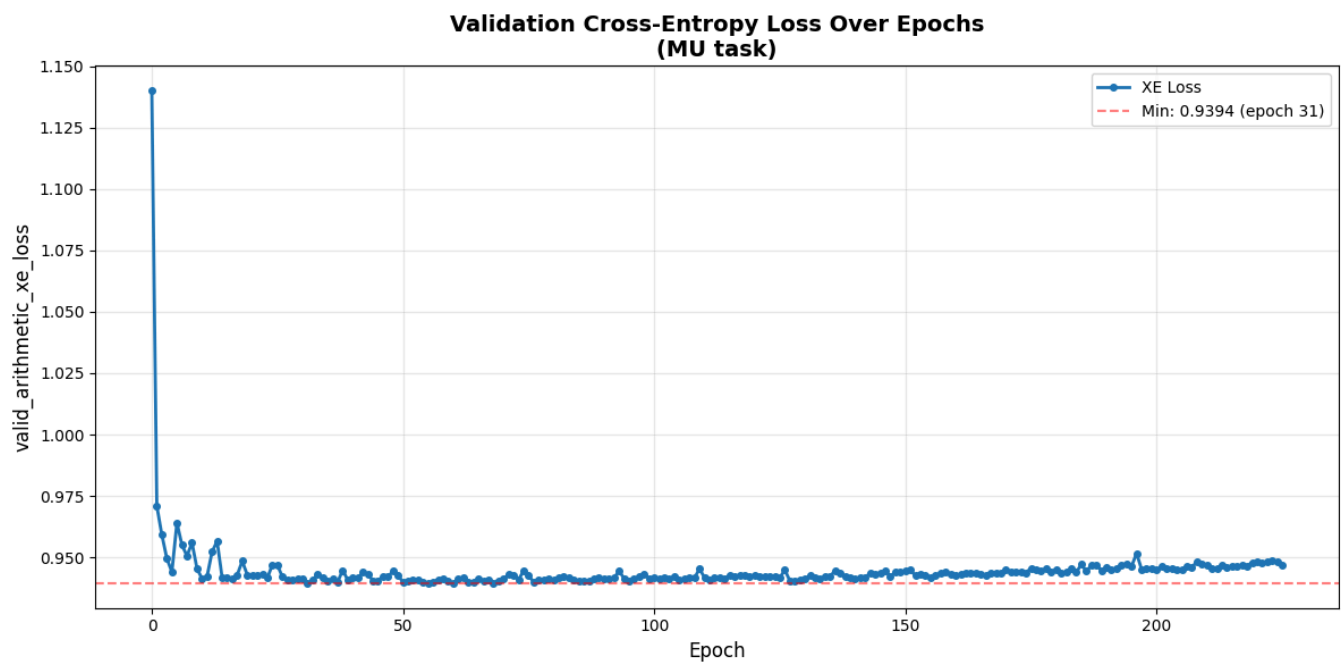
Convert Metrics to DataFrame

Total epochs recorded: 226

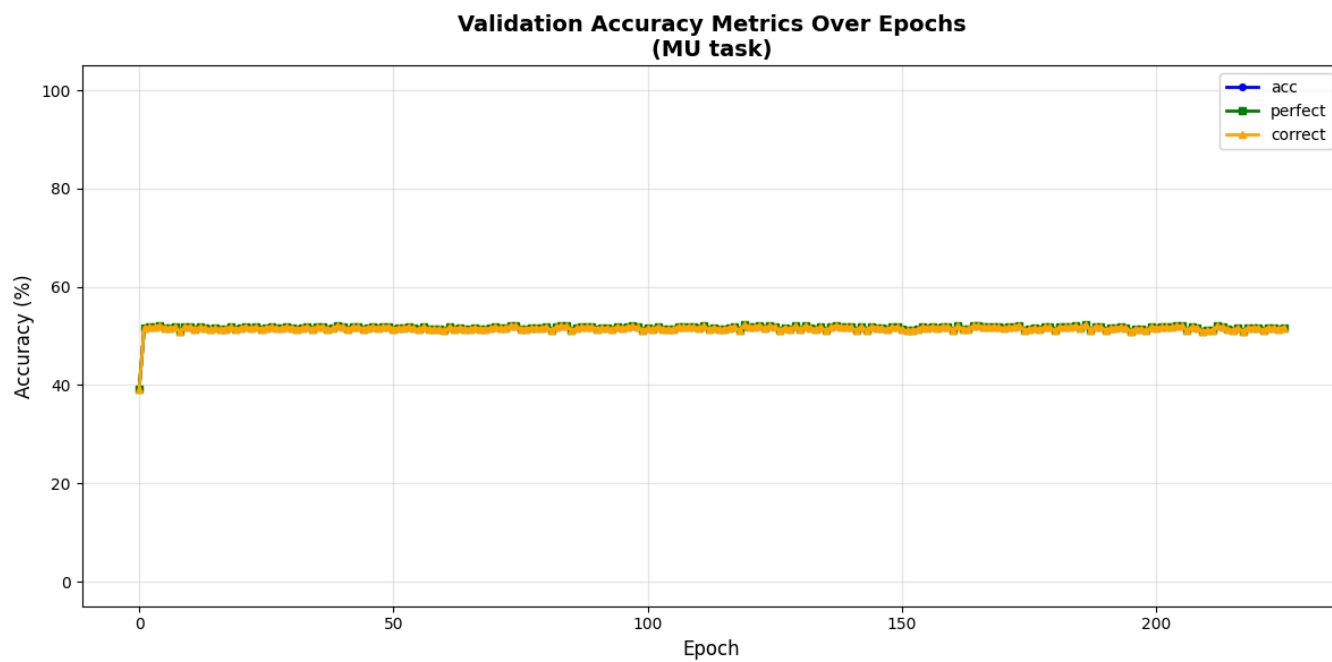
First few epochs:

	epoch	valid_arithmetic_xe_loss	valid_arithmetic_acc	valid_arithmetic_perfect	valid_arithmeti
0	0	1.140077	39.17	39.17	
1	1	0.970928	51.72	51.72	
2	2	0.959174	51.78	51.78	
3	3	0.949501	51.88	51.88	
4	4	0.944101	52.09	52.09	
5	5	0.964060	51.69	51.69	
6	6	0.955055	51.53	51.53	
7	7	0.950586	51.83	51.83	
8	8	0.955974	50.85	50.85	
9	9	0.945418	51.95	51.95	

Plot 1: Cross-Entropy Loss

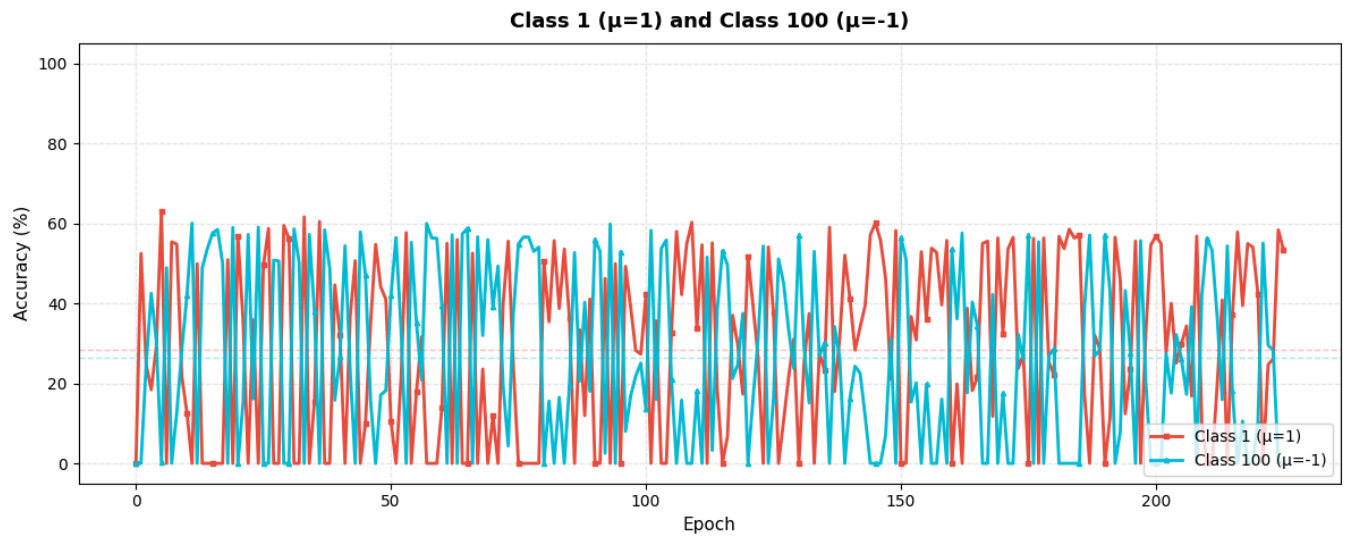
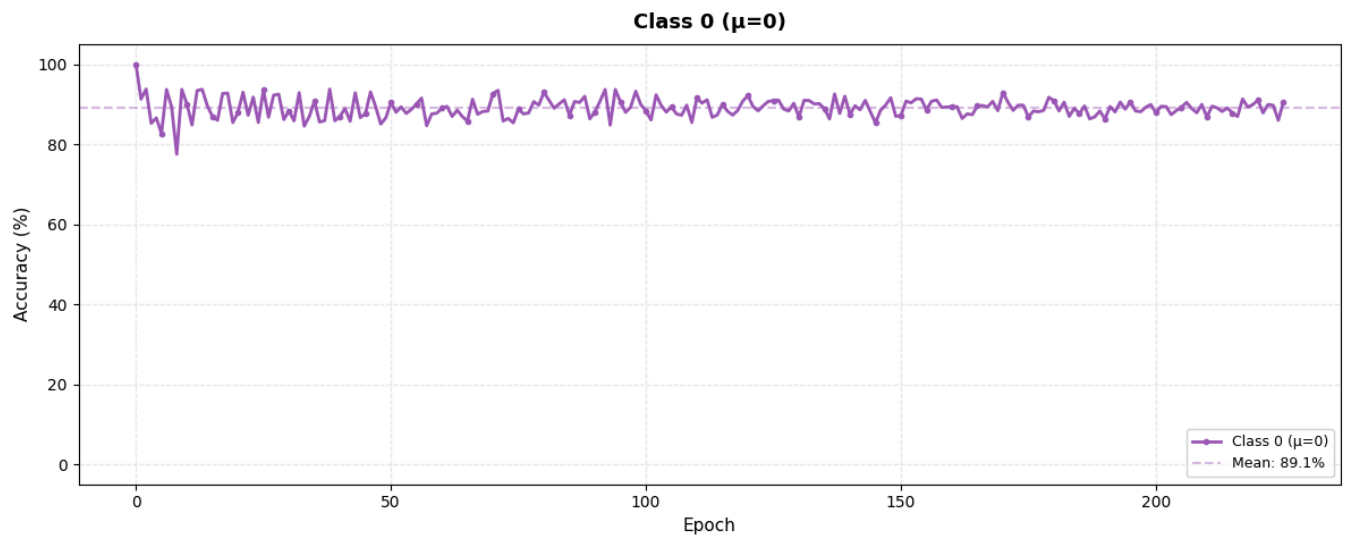


Plot 2: Accuracy Metrics (acc, perfect, correct)



Plot 3: Class-specific Accuracies (acc_0, acc_1, acc_100)

Class-Specific Accuracies Over Epochs (MU task)



Best Metrics Report

BEST METRICS ACROSS ALL EPOCHS

 CROSS-ENTROPY LOSS
Best Loss: 0.939445
Achieved at Epoch: 31

 ACCURACY METRICS

ACC:
Best: 52.24%
Achieved at Epoch: 186

PERFECT:
Best: 52.24%
Achieved at Epoch: 186

CORRECT:
Best: 52.24%
Achieved at Epoch: 186

 CLASS-SPECIFIC ACCURACIES

Class 0 ($\mu=0$):
Best: 100.00%
Achieved at Epoch: 0

Class 1 ($\mu=1$):
Best: 63.05%
Achieved at Epoch: 5

Class 100 ($\mu=-1$):
Best: 60.04%
Achieved at Epoch: 11

 LATEST EPOCH METRICS:
Epoch: 225.0
XE Loss: 0.946780
Accuracy: 51.72%
Perfect: 51.72%
